Curriculum Vitae

Mingsong Li

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Research Interests: Paleoclimatology; Paleohydrology; Sedimentology; Earth System Modeling; Data Assimilation; Time-Series Analysis; Software Development

1. Education

2016 Ph.D., China University of Geosciences (Wuhan), Geology
2014–2016 Johns Hopkins University (Visiting Ph.D. Student, CSC Fellow)
2012 M.S., Jilin University, China, Paleontology and Stratigraphy
2009 B.S., Jilin University, China, Geology

2. Professional Experience

Assistant Professor, Peking University (2020–present)
Assistant Research Professor, Pennsylvania State University (2019–2020)
Postdoctoral Researcher, Pennsylvania State University (2017–2018)
Postdoctoral Researcher, George Mason University (2016–2017)
Research Assistant, School of Earth Sciences, China University of Geosciences (2012–2016)
Research Assistant, College of Earth Sciences, Jilin University (2009–2012)

3. Selected Awards and Honors

2026 Sun Shu Prize, International Association of Sedimentologists (IAS)
2025 The 4th "China Sun Shu Award"
2025 Stanford/Elsevier World's Top 2% Scientists
2025 China's Top 10 Advances in Paleontology (2024)
2024 Stanford/Elsevier World's Top 2% Scientists
2024 Outstanding Editorial Group Member, National Science Review
2021 Peking University Boya Young Fellow
2021 AGU's Outstanding Reviewers
2019 Top 1% Peer Reviewer Awards, Publons, Web of Science Group

4. Research Funding

Projects Awarded Since 2021: RMB 9,519,300 in total funding (USD 1.42 million)

7. January 2025 – December 2027, National Natural Science Foundation of China, Original Exploration Program, "Revealing Carbon Release Sources and Processes of the Paleocene–Eocene Thermal Maximum through Cross-Sphere Isotope Data Assimilation (Main Application)" (42450195), **PI**, RMB 1,741,500
6. January 2025 – December 2028, National Natural Science Foundation of China, "Cyclostratigraphy of the secular resonance transition event of planetary orbits in the early Cenozoic chaotic Solar System" (42472050), **PI**, RMB 689,000
5. December 2022 – November 2027, National Key R&D Program of China (2022YFF0802900), "Temporal and spatial evolution of orbital scale ocean deoxygenation during the Paleocene-Eocene Thermal Maximum", **PI**, RMB 4,000,000

4. November 2021 – October 2026, National Key R&D Program of China (2021YFA0718200), "Metal stable isotope geochemical technique for tracing the evolution of the Earth's habitability in the middle Proterozoic", **Co-PI**, RMB 1,333,300
3. January 2021 – December 2024, National Natural Science Foundation of China (No. 42072040), "Study on the mechanism of lake level and sea level changes in the Early Triassic of China and Germany based on sedimentary noise modeling", **PI**, RMB 725,500
2. January 2021 – December 2022, the Fundamental Research Funds for the Central Universities (No. 7100603368), "High resolution astrochronology and paleoclimate change", **PI**, RMB 1,000,000
1. July 2021 – June 2022, Hubei Key Laboratory of Critical Zone Evolution, China University of Geosciences, Wuhan (No. 2021F07), "Sea level reconstruction from the Lower Triassic Xiejiacao section of Guangan, Sichuan", **PI**, RMB 30,000

Earlier Selected Research Support

3. March 2017 – December 2020, Heising-Simons Foundation award (No. 2016-011), "Paleoclimate Data Assimilation for Deep Time", Participant
2. September 2016 – February 2017, NSF-Standard Grant-OCE-1303605, "Collaborative Research: The relationship between multi-year droughts in California, coupled ocean-atmosphere climate oscillations and climate forcing", Participant
1. September 2014 – August 2016, China Scholarship Council (Grant No. 201406410029), "Astrochronology of Triassic Type Stratigraphic Sections in South China", PI

5. Software Copyrights and Patent Application

3. **Mingsong Li**. Time Series Analysis Software V2.8. Computer Software Copyright, Registration No. 2026SR0541650, **2026**.
2. Meng Wang, **Mingsong Li**. AcyclePy Time Series and Earth Science Data Analysis Software V1.0. Computer Software Copyright, Registration No. 2026SR0970480, **2026**.

1. Jiahao Liu, **Mingsong Li***, Xinwen Zhang. A Paleoclimate Reconstruction Method Based on Isotope Data Assimilation and Related Devices [Invention Patent Application]. Application No. 202511982369.1; Document Issue No. 2026033100373820. Applicant: Peking University, **2025**.

6. Research Software and Intellectual Property

Acycle: Time-series analysis software for paleoclimate research and education
https://github.com/mingsongli/acycle or https://acycle.org
A comprehensive and easy-to-use desktop application for macOS and Windows
Supporting the research of more than 1,000 scientists at institutions including the University of Cambridge, Yale University, Princeton University, Columbia University, the University of Calgary, the Australian National University, the Chinese Academy of Sciences, Utrecht University, the University of Padova, the University of Zaragoza, and the National Observatory (Brazil).

DeepDA: Paleoclimate data assimilation software
http://github.com/mingsongli/deepda

7. Publications

Google Scholar: https://scholar.google.com/citations?user=NYUXDW0AAAAJ
Oct. 1, 2026: h-index: 29; citations: 4,600
* = *Corresponding author*; # = *Student advisee*

In revision or under review:

Kaixuan Ji, Xiaoyu Zhang, Zhixin Wang, Fanhao Gong, Huaichun Wu, Laiming Zhang, Ren Wei, Yuan Gao, Zhijun Jin, **Mingsong Li***. 2026. Chaotic Earth–Mars resonance drove sea level and climate–biodiversity dynamics in the latest Cretaceous. ***Nature Communications***, in revision.

Jiahao Liu, Ruoyuan Qiu, Andy Ridgwell, Dan Zhu, Weiqi Yao, Liao Chang, Xinwen Zhang, Wang Miao, Ren Wei, Jianghui Du, Yongyun Hu, Zhijun Jin, **Mingsong Li***. 2026. Thermogenic carbon release reshaped the carbon cycle during the Paleocene–Eocene Thermal Maximum. ***Nature Communications***, under review.

Zhixin Wang, Yang Liu, Feng Ding, Kaixuan Ji, Jiahao Liu, Ren Wei, Haotian Zhang, Hanyu Zhu, Zhijun Jin, **Mingsong Li***. 2026. Orbital modulation of nonlinear climate dynamics on early Mars. ***Nature Astronomy***, under review.

Xinwen Zhang, Wanyi Lu, Babette Hoogakker, Weiqi Yao, Sifan Gu, Jiahao Liu, **Mingsong Li***. 2026. Reconstructing past bottom-water oxygen from benthic foraminiferal carbon-isotope offsets ($\Delta\delta^{13}C$): The BayDelC Bayesian framework. ***Paleoceanography and Paleoclimatology***, under review.

Mingsong Li*. 2026. Statistical tuning in cyclostratigraphy and astrochronology: foundations, validation, and methodological advances. ***Earth-Science Reviews***, under review.

Ren Wei, **Mingsong Li***, Xiaomei Wang*, Lin Dong, Yongyun Hu, Jian Cao, Meng Wang, Bo Jiu, Kaixuan Ji, Haotian Zhang, Zhijun Jin*. 2026. Orbitally gated monsoon wetlands as pulsed carbon sinks during the late Paleozoic Ice Age. ***Proceedings of the National Academy of Sciences***, under review.

Hongliang Lu, Baochun Huang, **Mingsong Li**, Qian Zhao, Xiang Li, Yilian Liao, Zhaowen Pei, Yi Xue, Josep M. Parés, Haotian Zhang, Qingqing Jiang. 2026. Synchronous End-Permian Mass Extinction Zenith in Marine and Low-Middle Latitude Land. ***Proceedings of the National Academy of Sciences***, in revision.

Tan Zhang, Chao Ma, Chao Li, Tailiang Fan, Chunju Huang, Anne-Christine Da Silva, Yifan Li, Qi Gao, He Huang, Meng Cheng, Yu Sui, **Mingsong Li**, Qiuyu Wang, Jiashan Lin, Hanting Zhong, Mingcai Hou, Chengshan Wang. 2026. Diachronous onset and comparatively clustered termination of the Ediacaran Shuram Excursion. ***Nature Communications***, under review.

Published or accepted:

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85. Meng Wang, Yukun Shi, Aihua Yang, Haotian Zhang, Xiaoyu Zhang, Yujing Wu, Yujie Shi, Jianbo Liu, Ren Wei, Zhijun Jin, **Mingsong Li***. 2026. Coupled sea-level and biodiversity dynamics during the Cambrian SPICE event. ***Nature Communications***, accepted.

84. Xiuwen Zhou, Ruiling Zhang, Man-Yin Tsang, **Mingsong Li**, Weiqi Yao. **2026**. Coupled increases in pelagic fish productivity and export productivity during the Paleocene-Eocene Thermal Maximum. ***Nature Communications***, 17(1): 10030. https://doi.org/10.1038/s41467-026-77863-9.

83. Gui-Mei Lu, Zhong-Shan Shen, Cheng-Long Deng, James G. Ogg, Ming-Dao Sun, **Mingsong Li**, Hua-Feng Qin, Ri-Xiang Zhu, Zhong-He Zhou, Yi-Gang Xu. **2026**. Refined chronologies of magnetochron M0r reveal asynchronous terrestrial and marine carbon isotope responses to Oceanic Anoxic Event 1a. ***Science Advances***, 12: eaea8374.

82. Qingqing Jiang, **Mingsong Li***, N. M. Papadomanolaki, Yonggang Liu, David De Vleeschouwer*. **2026**. Precession-related variability in North Pacific deep water formation in the Early Eocene: Insights from cGENIE simulations. ***Paleoceanography and Paleoclimatology***, 41: e2026PA005598.

81. Xiyue Liang, Dongjie Tang, Longfei Sun, Xiaoying Shi, **Mingsong Li**, Kunfeng Qiu, Baozeng Xie, Xinqiang Wang, Simon W. Poulton. **2026**. Sunspot cycles drove decadal fluctuations in shallow water oxygenation in the Mesoproterozoic. ***Earth and Planetary Science Letters***, 692: 120268.

80. Yilin Liu, Brandon B. Bell, Elizabeth M. Griffith, **Mingsong Li**, Yi Ge Zhang, Qingqing Jiang, Weiqi Yao. **2026**. Decoupling of biological pump strength and efficiency in the eastern equatorial Pacific during the Middle Miocene Climate Optimum. ***Earth and Planetary Science Letters***, 692: 120256.

79. Ren Wei, Zhijun Jin*, **Mingsong Li***, Yongyun Hu, Xiaomei Wang, Shuai Yuan, Meng Wang, Renda Huang, Haotian Zhang. **2026**. Orbital modulation of meridional atmospheric heat transport and ITCZ variability during the Late Paleozoic Ice Age. ***Journal of Geophysical Research: Atmospheres***, 131: e2026JD046636.

78. Zhijun Jin, Xiaomei Wang, Ren Wei, **Mingsong Li**, Yongyun Hu. **2026**. Multiscale hydrogeologic evolution and sedimentary responses during the Late Paleozoic Ice Age. ***Earth-Science Reviews***, 282: 105675.

77. Renda Huang, Zhijun Jin, Ruoyuan Qiu, Xueyu Yao, Yinan Feng, Ren Wei, **Mingsong Li**, Youwei Wang. **2026**. Obliquity modulation pacing of mid–high-latitude lacustrine systems during the Late Paleozoic Ice Age. ***Global and Planetary Change***, 264: 105554.

76. Ren Wei, Zhijun Jin*, **Mingsong Li***. **2026**. Geological records reveal chaotic diffusion during the Late Paleozoic Ice Age. ***Fundamental Research***, https://doi.org/10.1016/j.fmre.2026.03.023.

75. Naihua Xue, Wei Wang, David De Vleeschouwer, Chengguo Guan, **Mingsong Li**, Meng Wang, Xunlai Yuan, Philippe Claeys. **2026**. Astrochronology of the Ediacaran Period reveals the temporal heterogeneity and sea-level pacing of Shuram excursion. ***Precambrian Research***, 445: 108222.

74. Frits Hilgen, Jacques Laskar, **Mingsong Li**, Thomas Westerhold. **2026**. Cyclostratigraphy and astrochronology: The ultimate timekeepers of Earth's history. ***PAGES Magazine***, 34(1): 16–17.

73. **Mingsong Li**, Weijie Zhao. **2026**. Exploring the chaotic future and the deterministic past of the Solar System: An interview with Jacques Laskar. ***National Science Review***, 13(4): nwag002. https://doi.org/10.1093/nsr/nwag002.

72. **Mingsong Li***. **2026**. Acycle: Enhanced time-series analysis software for geoscience research and education. ***Science China Earth Sciences***, 69(3): 1066–1071. https://doi.org/10.1007/s11430-025-1848-7.

Mingsong Li*. **2026**. Acycle: Enhanced time-series analysis software for geoscience research and education. ***Scientia Sinica Terrae***, 56(3): 1100–1105. https://doi.org/10.1360/N072025-0527.

71. Haotian Zhang, Yan Chen, Paul B. Wignall, Kaixuan Ji*, Meng Wang, Xiaoyu Zhang, Hanyu Zhu, Ren Wei, Zhijun Jin, **Mingsong Li***. **2026**. Orbital forcing of paleoclimate and sedimentation in the Early Triassic: Insights from the Dayulin Section, North China. ***Science China Earth Sciences***, 69(9): 3306–3320. https://doi.org/10.1007/s11430-026-1953-3.

Haotian Zhang, Yan Chen, Paul B. Wignall, Kaixuan Ji*, Meng Wang, Xiaoyu Zhang, Hanyu Zhu, Ren Wei, Zhijun Jin, **Mingsong Li***. **2026**. Orbital forcing of paleoclimate and sedimentation in the Early Triassic: Insights from the Dayulin Section, North China. ***Scientia Sinica Terrae***, 56(9): 3355–3370. https://doi.org/10.1360/N072026-0067.

70. Qian He, Haotian Zhang, Stephen P. Hesselbo, Sofie Lindström, Sev Kender, Christian J. Bjerrum, **Mingsong Li***, Jianxin Yu, Yongdong Wang, Thomas J. Algeo, Yong Lei, Jun Shen*. **2026**. Astronomically tuned chronostratigraphy of an Upper Triassic–Middle Jurassic lacustrine succession in Southwest China. ***Global and Planetary Change***, 259: 105325. https://doi.org/10.1016/j.gloplacha.2026.105325.

69. Ke Xu, Nan Wu, Jianye Ren, David B. Kemp, Lin Ye, Zhongheng Sun, Xiangrong Yang, Xianghui Zhang, **Mingsong Li**. **2026**. Orbital forcing across a predominantly eolian record of the Eocene–Oligocene transition, Inner Mongolia, China. ***Global and Planetary Change***, 257: 105247. https://doi.org/10.1016/j.gloplacha.2025.105247.

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68. Haotian Zhang, Yan Chen, James G. Ogg, Zhiming Sun, Paul B. Wignall, Meng Wang, Haoxun Zhang, Xiaoyu Zhang, Yang Zhang, Keke Huang, Hongliang Lu, Baochun Huang, **Mingsong Li***. **2025**. Astronomically calibrated integrated stratigraphy of the Induan Stage (Early Triassic) and

significance for the Permian–Triassic mass extinction and aftermath. ***Earth and Planetary Science Letters***, 669: 119563.

67. Qingqing Jiang, **Mingsong Li***, Weiqi Yao, Ren Wei, Kaixuan Ji, Haotian Zhang, Zhijun Jin. **2025**. Astrochronology of the Paleocene–Eocene Thermal Maximum on the East Tasman Plateau. ***Global and Planetary Change***, 252: 104882. https://doi.org/10.1016/j.gloplacha.2025.104882.

66. Ren Wei, Zhijun Jin, **Mingsong Li**, Shuai Yuan, Yongyun Hu, Lin Dong, Rui Zhang, Juye Shi. **2025**. Tectonic–astronomical interactions in shaping late Paleozoic climate and organic carbon burial. ***Nature Communications***, 16: 8805.

65. Tan Zhang, Chao Ma, Yifan Li, Chao Li, Anne-Christine Da Silva, Tailiang Fan, Qi Gao, Mingzhi Kuang, Wangwei Liu, **Mingsong Li**, Mingcai Hou. **2025**. Astronomically calibrating early Ediacaran evolution. ***Nature Communications***, 16: 3049.

64. Renda Huang, **Mingsong Li**, Ruoyuan Qiu, Youwei Wang, Rui Zhang, Zhijun Jin. **2025**. Enhancing astronomical signal detection in cyclostratigraphy by superimposed averaging of paleoclimate proxies. ***Paleoceanography and Paleoclimatology***, 40: e2025PA005135.

63. Haoxun Zhang, **Mingsong Li***. **2025**. Reconstruction of global sea surface temperatures during the Early Eocene Climatic Optimum using paleoclimate data assimilation. ***Geological Journal of China Universities***, 31(3): 275–285.

62. Haoxun Zhang, **Mingsong Li***, Yongyun Hu. **2025**. Paleoclimate data assimilation: Principles and prospects. ***Science China Earth Sciences***, 68(2): 407–424. https://doi.org/10.1007/s11430-024-1439-y.

Haoxun Zhang, **Mingsong Li***, Yongyun Hu. **2025**. Paleoclimate data assimilation: Principles and prospects. ***Scientia Sinica Terrae***, https://doi.org/10.1360/SSTe-2024-0090.

61. Kai Wang, **Mingsong Li**, Peng Tang, Junxuan Fan, Renbin Zhan, Jianbo Liu. **2025**. High-resolution sedimentary cyclostratigraphy and astronomical signals in the Upper Ordovician of Southwest China. ***Palaeoworld***, 34(3): 100893. https://doi.org/10.1016/j.palwor.2024.11.001.

2024

60. **Mingsong Li***, Lee R. Kump, Andy Ridgwell, Jessica E. Tierney, Gregory J. Hakim, Steven B. Malevich, Christopher J. Poulsen, Robert Tardif, Haoxun Zhang, Jiang Zhu. **2024**. Coupled decline in ocean pH and carbonate saturation during the Palaeocene–Eocene Thermal Maximum. ***Nature Geoscience***, 17: 1299–1305.

59. **Mingsong Li***, Weiqi Yao, Jun Shen, Yuangeng Huang, Yi Yang. **2024**. Orbital-scale spatiotemporal dynamics of ocean deoxygenation during Paleocene-Eocene Thermal Maximum. ***China Basic Science***, 26: 24–30.

58. Meng Wang, **Mingsong Li***, Elizabeth A. Hajek, David B. Kemp, Yujing Wu, Hanyu Zhu, Chunju Huang, Haotian Zhang, Kaixuan Ji, Rui Zhang, Ren Wei, Zhijun Jin*. **2024**. Assessing the preservation of orbital signals across different sedimentary environments: Insights from stochastic sedimentation modeling. ***Earth and Planetary Science Letters***, 642: 118866.

57. Kai Wang, **Mingsong Li**, Aonan Zheng, Rongchang Wu, Xin Wei, Jianbo Liu. **2024**. Comprehensive cyclostratigraphy of the Katian Stage (Upper Ordovician) at the Wawenlatu section in Sichuan Province, Southwest China. ***Journal of Stratigraphy***, 48: 253–278.

56. Pingping Liu, Jiamin Lin, Dianbing Wang, Ben Ma, **Mingsong Li**. 2024. A tentative study on patterns of volcanic activities and their influencing factors. ***Bulletin of Mineralogy, Petrology and Geochemistry***, 43(5): 912–920. https://doi.org/10.3724/j.issn.1007-2802.20240038.

55. Xu Yao, Shuang Dai, Zhixiang Wang, **Mingsong Li**, Linda Hinnov. **2024**. Astronomical influence on Miocene continental environments in Tibet, east-central Asia. ***Geological Society of America Bulletin***, 136(7–8): 3457–3471. https://doi.org/10.1130/B37129.1.

2023

54. **Mingsong Li***, Haotian Zhang, Meng Wang, Zhijun Jin. **2023**. Astronomically forced changes in groundwater reservoirs and sea level during the greenhouse world. ***Chinese Science Bulletin***, 68(12): 1517–1527. https://doi.org/10.1360/TB-2022-0790.

53. **Meng Wang**, **Mingsong Li***, David B. Kemp, Jan Landwehrs, Zhijun Jin. **2023**. Late Triassic sedimentary records reveal the hydrological response to climate forcing and the history of the chaotic Solar System. ***Earth and Planetary Science Letters***, 607: 118052. https://doi.org/10.1016/j.epsl.2023.118052.

52. Braun Natália Santos, Ciro Climaco Rodrigues, Daniel Ribeiro Franco, **Mingsong Li**, Mariana Aragão Fernandes, Mariane Candido, Raysa de Magalhães Rocha, Thiago Pereira dos Santos, André De Gasperi, Daniel Galvão Carnier Fragoso, Gabriella Fazio, Ana Natalia Gomes Rodrigues. 2023. Integration of seismic stratigraphy and cyclostratigraphy for high resolution chronostratigraphic correlation: The Albacora Field, Campos Basin, Brazil. ***Marine and Petroleum Geology***, 158: 106541. https://doi.org/10.1016/j.marpetgeo.2023.106541.

51. Dongyang Liu, Chunju Huang, **Mingsong Li**, Wenhan Chen, Runjian Chu. **2023**. Evolution of phosphorus enrichment in the early to middle Middle Triassic Epoch in Southern China's ancient oceans and its response to astronomical forcing. ***Quaternary Sciences***, 43(6): 1538–1546.

50. Meng Wang, Qing Li, Yanguang Dou, Feng Cai, **Mingsong Li**. **2023**. Astronomically forced climate changes in the Okinawa Trough since the Late Pleistocene: Evidence from CSHC-15 core. ***Quaternary Sciences***, 43(6): 1711–1721.

49. Rui Zhang, Zhijun Jin, **Mingsong Li**, Quanyou Liu, Peng Li, Ren Wei, Meng Wang, Xiangwu He. **2023**. Astronomical regulation of the Middle Triassic organic-rich shales in the Ordos Basin. ***Quaternary Sciences***, 43(6): 1547–1561.

48. Rui Zhang, Zhijun Jin, Rukai Zhu, **Mingsong Li**, Xiao Hui, Ren Wei, Xiangwu He, Qian Zhang. **2023**. Investigation of deposition rate of terrestrial organic-rich shales in China and its implications for shale oil exploration. ***Oil & Gas Geology***, 44(4): 829–845.

47. Yiquan Ma, Majie Fan, **Mingsong Li***, James G. Ogg, Chen Zhang, Jun Feng, Chunhua Zhou, Xiaofeng Liu, Yongchao Lu, Huimin Liu, James S. Eldrett, Chao Ma*. **2023**. East Asian lake hydrology modulated by global sea-level variations in the Eocene warmhouse. ***Earth and Planetary Science Letters***, 602: 117925. https://doi.org/10.1016/j.epsl.2022.117925.

46. Ren Wei, Zhijun Jin, Rui Zhang, **Mingsong Li**, Yongyun Hu, Xiangwu He, Shuai Yuan. **2023**. Orbitally-paced coastal sedimentary records and global sea-level changes in the early Permian. ***Earth and Planetary Science Letters***, 620: 118356. https://doi.org/10.1016/j.epsl.2023.118356.

45. Rui Zhang, David B. Kemp, Nicolas Thibault, Mads E. Jelby, **Mingsong Li**, Chunju Huang, Yu Sui, Zhixiang Wang, Dongyang Liu, Shizhen Jia. **2023**. Astrochronology and sedimentary noise modeling of Pliensbachian (Early Jurassic) sea-level changes, Paris Basin, France. ***Earth and Planetary Science Letters***, 614: 118199. https://doi.org/10.1016/j.epsl.2023.118199.

44. Rui Zhang, Zhijun Jin, **Mingsong Li**, Michael Gillman, Shuping Chen, Quanyou Liu, Ren Wei, Juye Shi. **2023**. Long-term periodicity of sedimentary basins in response to astronomical forcing: Review and perspective. ***Earth-Science Reviews***, 244: 104533. https://doi.org/10.1016/j.earscirev.2023.104533.

43. Chun-Sheng Jin, Deke Xu, **Mingsong Li**, Pengxiang Hu, Zhaoxia Jiang, Jianxing Liu, Yunfa Miao, Fuli Wu, Wentian Liang, Qiang Zhang, Bai Su, Qingsong Liu, Ran Zhang, Jimin Sun. **2023**. Tectonic and orbital forcing of the South Asian monsoon in central Tibet during the late Oligocene. ***Proceedings of the National Academy of Sciences***, 120(15): e2214558120. https://doi.org/10.1073/pnas.2214558120.

42. Ren Wei, **Mingsong Li**, Rui Zhang, Yongyun Hu, James G. Ogg, Guoyong Liu, He Huang, Xiangwu He, Shuai Yuan, Qifan Lin, Zhijun Jin. **2023**. Obliquity forcing of continental aquifers during the late Paleozoic ice age. ***Earth and Planetary Science Letters***, 613: 118174. https://doi.org/10.1016/j.epsl.2023.118174.

41. Ren Wei, Rui Zhang, **Mingsong Li**, Xiaojun Wang, Zhijun Jin. **2023**. Obliquity forcing of lake-level changes and organic carbon burial during the Late Paleozoic Ice Age. ***Global and Planetary Change***, 223: 104092. https://doi.org/10.1016/j.gloplacha.2023.104092.

40. Ruoyuan Qiu, Linhao Fang, Yuanzheng Lu, Yuxuan Chen, Renda Huang, Wenzhi Lei, Pengyuan Zhang, **Mingsong Li**. **2023**. Cyclostratigraphy of the Lower Jurassic (Toarcian) terrestrial successions in the Sichuan Basin, southwestern China. ***Journal of Asian Earth Sciences***, 250: 105617. https://doi.org/10.1016/j.jseaes.2023.105617.

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39. **Mingsong Li***, Timothy J. Bralower, Lee R. Kump, Jean M. Self-Trail, James C. Zachos, William D. Rush, Marci M. Robinson. **2022**. Astrochronology of the Paleocene-Eocene Thermal Maximum on the Atlantic Coastal Plain. ***Nature Communications***, 13(1): 5618. https://doi.org/10.1038/s41467-022-33390-x.

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37. Jun Tian, Huaichun Wu, Chunju Huang, **Mingsong Li**, Chao Ma, Pinxian Wang. **2022**. Revisiting the Milankovitch Theory from the Perspective of the 405 ka Long Eccentricity Cycle. ***Earth Science***, 47(10): 3543–3568. https://doi.org/10.3799/dqkx.2022.248 (Chinese).

36. Jessica E. Tierney, Jiang Zhu, **Mingsong Li**, Andy Ridgwell, Gregory J. Hakim, Christopher J. Poulsen, Ross D. M. Whiteford, James W. B. Rae, Lee R. Kump. **2022**. Spatial patterns of climate change across the Paleocene–Eocene Thermal Maximum. ***Proceedings of the National Academy of Sciences***, 119(42): e2205326119. https://doi.org/10.1073/pnas.2205326119.

35. Zhifeng Zhang, Yongjian Huang, **Mingsong Li**, Xiang Li, Pengcheng Ju, Chengshan Wang. **2022**. Obliquity-forced aquifer-eustasy during the Late Cretaceous greenhouse world. ***Earth and Planetary Science Letters***, 596: 117800. https://doi.org/10.1016/j.epsl.2022.117800.

34. **Yujing Wu**, Xianjun Fang, Linhua Jiang, Biao Song, Baofu Han, **Mingsong Li***, Jianqing Ji*. **2022**. Very long-term periodicity of episodic zircon production and Earth system evolution. ***Earth-Science Reviews***, 233: 104164. https://doi.org/10.1016/j.earscirev.2022.104164.

33. Xu Yao, Shuang Dai, **Mingsong Li**, Linda Hinnov. **2022**. Orbital eccentricity and inclination metronomes in Middle Miocene lacustrine mudstones of Jiuxi Basin, Tibet: Closing an astrochronology time gap and calibrating global cooling events. ***Global and Planetary Change***, 215: 103896. https://doi.org/10.1016/j.gloplacha.2022.103896.

32. Tan Zhang, Yifan Li*, Tailiang Fan, Anne-Christine Da Silva, Juye Shi, Qi Gao, Mingzhi Kuang, Wangwei Liu, Zhiqian Gao, **Mingsong Li***. 2022. Orbitally-paced climate change in the early Cambrian and its implications for the history of the Solar System. ***Earth and Planetary Science Letters***, 583: 117420. https://doi.org/10.1016/j.epsl.2022.117420.

31. Simin Jin, David B. Kemp, David W. Jolley, Manuel Vieira, James C. Zachos, Chunju Huang, **Mingsong Li**, Wenhan Chen. **2022**. Large-scale, astronomically paced sediment input to the North Sea Basin during the Paleocene Eocene Thermal Maximum. ***Earth and Planetary Science Letters***, 579: 117340. https://doi.org/10.1016/j.epsl.2021.117340.

30. **Meng Wang**, **Mingsong Li***, David B. Kemp, Slah Boulila, James G. Ogg. 2022. Sedimentary noise modeling of lake-level change in the Late Triassic Newark Basin of North America. ***Global and Planetary Change***, 208: 103706. https://doi.org/10.1016/j.gloplacha.2021.103706.

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29. Ying Cui*, **Mingsong Li***, Elsbeth E. van Soelen, Francien Peterse, Wolfram M. Kürschner*. **2021**. Massive and rapid predominantly volcanic CO_2 emission during the end-Permian mass extinction. ***Proceedings of the National Academy of Sciences***, 118(37): e2014701118. https://doi.org/10.1073/pnas.2014701118.

28. Dongyang Liu, Chunju Huang, James G. Ogg, David B. Kemp, **Mingsong Li**, Meiyi Yu, William J. Foster. **2021**. Astronomically forced changes in chemical weathering and redox during the Anisian (Middle Triassic): Implications for marine ecosystem recovery following the end-Permian mass extinction. ***Palaeogeography, Palaeoclimatology, Palaeoecology***, 569: 110355. https://doi.org/10.1016/j.palaeo.2021.110355.

27. Dongyang Liu, Chunju Huang, David B. Kemp, **Mingsong Li**, James G. Ogg, Meiyi Yu, William J. Foster. **2021**. Paleoclimate and sea level response to orbital forcing in the Middle Triassic of the eastern Tethys. ***Global and Planetary Change***, 199: 103454. https://doi.org/10.1016/j.gloplacha.2021.103454.

26. Omid Falahatkhah, Ali Kadkhodaie*, Ali A. Ciabeghodsi, **Mingsong Li***. **2021**. Cyclostratigraphy of the Lower Triassic Kangan Formation in the Salman gas field, eastern Persian Gulf, Iran. ***Palaeogeography, Palaeoclimatology, Palaeoecology***, 561: 110045. https://doi.org/10.1016/j.palaeo.2020.110045.

2017–2020

25. Haocheng Yu, Kunfeng Qiu, **Mingsong Li**, M. Santosh, Z. G. Zhao, Y. Q. Huang. **2020**. Record of the late Paleozoic ice age from Tarim, China. ***Geochemistry, Geophysics, Geosystems***, 21(11): e2020GC009237. https://doi.org/10.1029/2020GC009237.

24. Zheng Gong, **Mingsong Li**. **2020**. Astrochronology of the Ediacaran Shuram carbon isotope excursion, Oman. ***Earth and Planetary Science Letters***, 547: 116462. https://doi.org/10.1016/j.epsl.2020.116462.

23. J. Fred Read, **Mingsong Li**, Linda A. Hinnov, Campbell S. Nelson, Steven Hood. **2020**. Testing for astronomical forcing of cycles and gamma ray signals in outer shelf/upper slope, mixed siliciclastic-carbonates: Upper Oligocene, New Zealand. ***Palaeogeography, Palaeoclimatology, Palaeoecology***, 555: 109821. https://doi.org/10.1016/j.palaeo.2020.109821.

22. Yang Zhang, James G. Ogg, Matthias Franz, Gerhard H. Bachmann, Michael Szurlies, Heinz-Gerd Röhling, **Mingsong Li**, Christian Rolf, Karsten Obst. **2020**. Carnian (Late Triassic) magnetostratigraphy from the Germanic Basin allowing global correlation of the Mid-Carnian Episode. ***Earth and Planetary Science Letters***, 541: 116275. https://doi.org/10.1016/j.epsl.2020.116275.

21. Chao Ma, **Mingsong Li**. **2020**. Astronomical time scale of the Turonian constrained by multiple paleoclimate proxies. ***Geoscience Frontiers***, 11: 1345–1352. https://doi.org/10.1016/j.gsf.2020.01.013.

20. **Meng Wang**, Honghan Chen*, Chunju Huang, David B. Kemp, Tianwu Xu, Hongan Zhang, **Mingsong Li***. **2020**. Astronomical forcing and sedimentary noise modeling of lake-level changes in the Paleogene Dongpu Depression of North China. ***Earth and Planetary Science Letters***, 535: 116116. https://doi.org/10.1016/j.epsl.2020.116116.

19. Tan Zhang, Changmin Zhang, Tailiang Fan, Lei Zhang, Rui Zhu, Jinyu Tao, **Mingsong Li**. **2020**. Cyclostratigraphy of Lower Triassic terrestrial successions in the Junggar Basin, northwestern China. ***Palaeogeography, Palaeoclimatology, Palaeoecology***, 539: 109493. https://doi.org/10.1016/j.palaeo.2019.109493.

18. **Mingsong Li**, Hubert L. Barnes. **2019**. Orbitally forced sphalerite growth in the Upper Mississippi Valley District. ***Geochemical Perspectives Letters,*** 12: 18–22. https://doi.org/10.7185/geochemlet.1929 [***Cover story***].

17. Peng Gao, Junsheng Nie, **Mingsong Li**, Pu Li. **2019**. Confirmation of a late Miocene subchron C4n.2n-1r from the eastern Qaidam Basin in the NE Tibetan Plateau. ***Journal of Geophysical Research: Solid Earth***, 124(12): 12354–12365. https://doi.org/10.1029/2019JB017936.

16. Matthias Sinnesael, David De Vleeschouwer, Christian Zeeden, Sietske J. Batenburg, Anne-Christine Da Silva, Niels J. de Winter, Jaume Dinarès-Turell, Anna Joy Drury, Gabriele Gambacorta,

Frederik J. Hilgen, Linda A. Hinnov, Alexander J. L. Hudson, David B. Kemp, Margriet L. Lantink, Jiří Laurin, **Mingsong Li**, Diederik Liebrand, Chao Ma, Stephen R. Meyers, Johannes Monkenbusch, Alessandro Montanari, Theresa Nohl, Heiko Pälike, Damien Pas, Micha Ruhl, Nicolas Thibault, Maximilian Vahlenkamp, Luis Valero, Sébastien Wouters, Huaichun Wu, Philippe Claeys. **2019**. The Cyclostratigraphy Intercomparison Project (CIP): consistency, merits and pitfalls. ***Earth-Science Reviews*****,** 199: 102965. https://doi.org/10.1016/j.earscirev.2019.102965.

15. Qingda Su, Junsheng Nie, Zeng Luo, **Mingsong Li**, Richard Heermance, Carmala Garzione. **2019**. Detection of strong precession cycles from the late Pliocene sedimentary records of northeastern Tibetan Plateau. ***Geochemistry, Geophysics, Geosystems***, 20(8): 3901–3912. https://doi.org/10.1029/2019GC008447.
14. **Mingsong Li**, Chunju Huang, James Ogg, Yang Zhang, Linda Hinnov, Huaichun Wu, Zhong-Qiang Chen, Zhuoyan Zou. **2019**. Paleoclimate proxies for cyclostratigraphy: Comparative analysis using a Lower Triassic marine section in South China. ***Earth-Science Reviews,*** 189: 125–146. https://doi.org/10.1016/j.earscirev.2019.01.011.
13. **Mingsong Li**, Linda Hinnov, Lee Kump. **2019**. *Acycle*: Time-series analysis software for paleoclimate research and education. ***Computers & Geosciences***, 127: 12–22. https://doi.org/10.1016/j.cageo.2019.02.011.
12. **Mingsong Li**, Linda A. Hinnov, Chunju Huang, James G. Ogg. **2018**. Sedimentary noise and sea levels linked to land–ocean water exchange and obliquity forcing. ***Nature Communications***, 9: 1004. https://doi.org/10.1038/s41467-018-03454-y.
11. **Mingsong Li**, Lee R. Kump, Linda A. Hinnov, Michael E. Mann. **2018**. Tracking variable sedimentation rates and astronomical forcing in Phanerozoic paleoclimate proxy series with evolutionary correlation coefficients and hypothesis testing. ***Earth and Planetary Science Letters***, 501: 165–179. https://doi.org/10.1016/j.epsl.2018.08.041.
10. **Mingsong Li**, Chunju Huang, Linda Hinnov, Weizhe Chen, James Ogg, Wei Tian. **2018**. Astrochronology of the Anisian stage (Middle Triassic) at the Guandao reference section, South China. ***Earth and Planetary Science Letters***, 482: 591–606. https://doi.org/10.1016/j.epsl.2017.11.042.
9. **Mingsong Li**, Yang Zhang, Chunju Huang, James Ogg, Linda Hinnov, Yongdong Wang, Zhuoyan Zou, Liqin Li. **2017**. Astronomical tuning and magnetostratigraphy of the Upper Triassic Xujiahe Formation of South China and Newark Supergroup of North America: Implications for the Late Triassic time scale. ***Earth and Planetary Science Letters***, 475: 207–223. https://doi.org/10.1016/j.epsl.2017.07.015.

2010–2016

8. **Mingsong Li**, Chunju Huang, Linda Hinnov, James Ogg, Zhong-Qiang Chen, Yang Zhang. **2016**. Obliquity-forced climate during the Early Triassic hothouse in China. ***Geology***, 44(8): 623–626. https://doi.org/10.1130/G37970.1 [***Cover story***].
7. **Mingsong Li**, James Ogg, Yang Zhang, Chunju Huang, Linda Hinnov, Zhong-Qiang Chen, Zhuoyan Zou. **2016**. Astronomical tuning of the end-Permian extinction and the Early Triassic Epoch of South China and Germany. ***Earth and Planetary Science Letters,*** 441: 10–25. https://doi.org/10.1016/j.epsl.2016.02.017.
6. Zhuoyan Zou, Chunju Huang, **Mingsong Li**, Yang Zhang. **2016**. Climate change response to astronomical forcing during the Oligocene-Miocene transition in the equatorial Atlantic (ODP Site 926). ***Science China Earth Sciences***, 59(8): 1665–1673. https://doi.org/10.1007/s11430-016-5311-y.
5. Yang Zhang, **Mingsong Li**, James G. Ogg, Paul Montgomery, Chunju Huang, Zhong-Qiang Chen, Zhiqiang Shi, Paul Enos, Daniel J. Lehrmann. **2015**. Cycle-calibrated magnetostratigraphy of middle Carnian from South China: Implications for Late Triassic time scale and termination of the Yangtze Platform. ***Palaeogeography, Palaeoclimatology, Palaeoecology***, 436: 135–166. https://doi.org/10.1016/j.palaeo.2015.05.033.

4. Yuewu Sun, **Mingsong Li**, Wenchun Ge, Yanlong Zhang, Dejun Zhang. **2013**. Eastward termination of the Solonker–Xar Moron River Suture determined by detrital zircon U–Pb isotopic dating and Permian floristics. ***Journal of Asian Earth Sciences***, 75: 243–250. https://doi.org/10.1016/j.jseaes.2013.07.018.
3. Yuewu Sun, **Mingsong Li,** Guowei Zhao. **2012**. A new terrestrial lithostratigraphic unit of the Cisuralian (Early Permian) in the Yanbian area, Jilin Province. ***Journal of Stratigraphy***, 36: 89–96.
2. **Mingsong Li**, Yuewu Sun, Guowei Zhao. **2011**. Discovery of Early Permian Cathaysia flora from Daxinggou of Wangqing County, Yanbian area, Jilin Province, China and its geological significance. ***Advances in Earth Science***, 26: 339–346.
1. Shuqin Zhang, Yuying Liu, **Mingsong Li**. **2010**. Eocene-Miocene palynological assemblages in Wanchang area of Jilin and their stratigraphic significance. ***Global Geology***, 29: 357–362.

8. Speaking Engagements

52. National Space Science Center, Chinese Academy of Sciences, Beijing, China, September 24, 2026. **Invited Speaker**.
51. Chongqing University of Science and Technology, Chongqing, China, May 13, 2026. **Invited Speaker**.
50. Peking University Library, Beijing, China, April 17, 2026. **Invited Speaker**.
49. Tongji University, Shanghai, China, January 26, 2026. **Invited Speaker**.
48. School of Atmospheric Sciences, Lanzhou University, Lanzhou, China, December 8, 2025. **Invited Speaker**.
47. School of Earth Sciences, Lanzhou University, Lanzhou, China, December 8, 2025. **Invited Speaker**.
46. Northwest University, Xi'an, China, December 1, 2025. **Invited Speaker**.
45. Shuangqing Forum, Beijing, China, November 24–25, 2025. **Invited Speaker**.
44. China University of Petroleum (East China), Qingdao, China, May 26, 2025. **Invited Speaker**.
43. Ocean University of China, Qingdao, China, May 27, 2025. **Invited Speaker**.
42. Tongji University, Shanghai, China, April 11–12, 2025.
41. Nanjing Institute of Geology and Palaeontology, Chinese Academy of Sciences, Nanjing, China, January 14, 2025. **Invited Speaker**.
40. China University of Petroleum (Beijing), Beijing, China, January 2, 2025. **Invited Speaker**.
39. 7th Young Scholars Forum of Palaeontology, Chengdu, China, December 16, 2024. **Invited Speaker**.
38. Institute of Geology and Geophysics, Chinese Academy of Sciences, Beijing, China, December 4, 2024. **Invited Speaker**.
37. Yuanpei College, Peking University, Beijing, China, November 15, 2024. **Invited Speaker**.
36. Paris Observatory, Paris, France, July 17, 2024. **Invited Speaker**.
35. Workshop: Cyclostratigraphy Intercomparison Project 2.0, Brussels, Belgium, July 9, 2024. **Keynote Speaker**.
34. Shenyang Normal University, Shenyang, China, June 18, 2024. **Invited Speaker**.
33. School of Earth and Space Sciences, Peking University, Beijing, China, March 8, 2024. **Invited Speaker**.
32. Institute of Atmospheric Physics, Chinese Academy of Sciences, Beijing, China, September 15, 2023. **Invited Speaker**.
31. Research Institute of Petroleum Exploration and Development, China National Petroleum Corporation, Beijing, China, April 8, 2023. **Invited Speaker**.
30. Institute of Tibetan Plateau Research, Chinese Academy of Sciences, Beijing, China, April 19, 2023. **Invited Speaker**.
29. Institute of Earth Environment, Chinese Academy of Sciences, "Legend of Past and Present" Forum, Beijing, China, April 13, 2023. **Invited Speaker**.
28. Paleoclimatology Group Seminar, Beijing, China, February 5, 2023. **Invited Speaker**.
27. Solid Earth Science Key Laboratory Alliance 2022 Academic Committee Meeting, Beijing, China, December 18, 2022. **Invited Speaker**.

26. National Natural Science Foundation of China, Continental Evolution and Monsoon System Evolution Basic Science Center Project, Online Academic Seminar Series, Beijing, China, October 15, 2022. **Invited Speaker**.
25. Jilin University, Changchun, China, May 9, 2022. **Invited Speaker**.
24. Cloud Meeting on Paleomagnetism, Beijing, China, December 5, 2021. **Invited Speaker**.
23. Peking University, Beijing, China, October 27, 2021. **Invited Speaker**.
22. Southern University of Science and Technology, Shenzhen, China, October 23, 2021. **Invited Speaker**.
21. Montclair State University, October 11, 2021. **Invited Speaker**.
20. Tongji University, Shanghai, China, October 11, 2021. **Invited Speaker**.
19. Peking University, Beijing, China, June 11, 2021.
18. Peking University, Beijing, China, May 26, 2021.
17. Tongji University, Shanghai, China, April 19, 2021. **Invited Speaker**.
16. China University of Geosciences (Beijing), Beijing, China, April 16, 2021. **Invited Speaker**.
15. Virginia Tech, Blacksburg, Virginia, USA, March 18, 2021. **Invited Speaker**.
14. Institute of Geology and Geophysics, Chinese Academy of Sciences, Beijing, China, March 16, 2021. **Invited LIU Tungsheng Lecture**.
13. Virginia Tech, Blacksburg, Virginia, USA, March 11, 2021. **Invited Speaker**.
12. Pennsylvania State University, State College, USA, February 14, 2020.
11. Lehigh University, Department of Earth and Environmental Sciences, Bethlehem, USA, September 5, 2019. **Invited Speaker**.
10. George Mason University, Department of Atmospheric, Oceanic and Earth Sciences, Geology Seminar Series, Fairfax, Virginia, USA, March 28, 2019. **Invited Speaker**.
9. Pennsylvania State University, State College, Pennsylvania, USA, September 21, 2018.
8. Nanjing Institute of Geology and Palaeontology, Chinese Academy of Sciences, Nanjing, Jiangsu, China, May 11, 2018. **Invited Speaker**.
7. Nanjing University, Nanjing, Jiangsu, China, May 9, 2018. **Invited Speaker**.
6. China University of Geosciences, Wuhan, Hubei, China, March 26, 2018. **Invited Speaker**.
5. Purdue University, West Lafayette, Indiana, USA, January 18, 2018. **Invited Speaker**.
4. Peking University, Beijing, China, October 19, 2017. **Invited Speaker**.
3. Pennsylvania State University, State College, Pennsylvania, USA, April 14, 2017.
2. China University of Geosciences, Wuhan, Hubei, China, March 19, 2017. **Invited Speaker**.
1. George Mason University, Department of Atmospheric, Oceanic and Earth Sciences, Geology Seminar Series, Fairfax, Virginia, USA, October 8, 2015.

9. Teaching Experience

Peking University

2022-2026, fall semesters. Deep-Time Earth Data Analysis and Visualization (01201270); Graduate, 2 credits. Instructor; taught in English in Fall 2026.

2023-2026, fall semesters. Data Science for Geoscience (01232010); Undergraduate, 2 credits. Instructor.

2024-2026, fall semesters. Seminar on Paleobiology and Stratigraphy (01217280); Graduate, 2 credits. Instructor; co-taught in 2025 and 2026.

August 2026. Integrated Field Course in Sedimentology, Stratigraphy, and Paleontology; Undergraduate, 2 credits. Co-taught; taught in English.

Spring 2021 and Spring 2022. Advances in Paleontology and Stratigraphy; graduate, 2 credits. Co-taught.

Earlier Teaching

Fall 2019. Interim Instructor, Pennsylvania State University. GEOSC 204, Geobiology; co-designed and taught lectures on paleoclimate and mass extinctions.

September 2019. Invited Short Course, Lehigh University. Acycle time-series analysis for paleoclimate research and education.

Spring 2016. Interim Instructor, George Mason University. GEOL 304, Sedimentary Geology; chronostratigraphic analysis using MATLAB and Excel. GEOL 565 / CLIM 759, Paleoceanography; fundamentals of geologic time.

2014. Teaching Assistant, China University of Geosciences (Wuhan). Astronomical Forcing of the Paleoclimate System; hands-on astrochronology and interpretation of chronostratigraphic information.

Workshop and Short Courses

April 2025, Acycle Cyclostratigraphy Workshop, Beijing, China. **Chair** (400+ participants).

April 2025, Short Course on Cyclostratigraphy, Sedimentological Congress, Beijing, China. Organizer and Lecturer.

April 2023, Short Course on Cyclostratigraphy, Sedimentological Congress, Chengdu University of Technology, Chengdu, China. Lecturer.

Other

July 2026, AI-Empowered Research and Education, Peking University, Beijing, China. **Invited Speaker**.

April 2026, The Logical Structure of High-Level Scientific Research: From Problem Awareness to Communicating Findings, Higher Education Press, Beijing, China. **Invited Speaker**.

January 2026, “Research Paper Writing in the AI Era”, Wuhan Business University, Wuhu, Anhui, China. **Invited Speaker**.

December 2024, Generative AI Revolutionizing Geoscience Research and Education, AI for Education Yanyuan Zhisheng Salon, Peking University, Beijing, China. **Invited Speaker**.

10. Student and Postdoctoral Mentorship

Current Postdoctoral and Graduate Researchers

Haotian Zhang: Boya Postdoctoral Fellow, since July 2026; supervisor
Xiaoyu Zhang: Ph.D. student, since September 2022; supervisor
Hanyu Zhu: Ph.D. student, since September 2023; supervisor
Zhixin Wang: Ph.D. student, since September 2024; supervisor
Xinwen Zhang: Ph.D. student, since September 2025; supervisor

Jiahao Liu: Master's student, since September 2026; supervisor

Completed Graduate and Postdoctoral Supervision

Haotian Zhang: Ph.D. student, 2021–2026; supervisor. Now a Boya Postdoctoral Fellow in the group.

Meng Wang: 2021–2023, Peking University Boya Postdoctoral Fellow; 2019–2020, visiting Ph.D. student at Penn State; currently an Associate Professor, Nanjing Institute of Geology and Palaeontology, Chinese Academy of Sciences (supported by the CAS Hundred Talents Program, Category B)

Kaixuan Ji: 2022–2026, Postdoctoral Researcher; currently an Assistant Professor, China University of Geosciences (Beijing)

Yujing Wu: 2021–2023, Ph.D. degree, associate supervisor; currently a postdoctoral researcher at IMCCE, Observatoire de Paris (advisor: Jacques Laskar)

Qingqing Jiang: Master's student, 2022–2025; supervisor; currently a Ph.D. student at the University of Münster

Haoxun Zhang: Master's student, 2021–2024; supervisor

Ciro Clímaco Rodrigues: 2021–2023, master's degree, National Observatory in Rio de Janeiro, Brazil; associate supervisor; currently at Petrobras

Mariana Aragão Fernandes: 2021–2023, master's degree, National Observatory in Rio de Janeiro, Brazil; associate supervisor

Undergraduate Thesis Supervisor

2023 Hanyu Zhu
2024 Zhixin Wang
2025 Xinwen Zhang
2026 Jiahao Liu
2027 Heng Wu (expected)
2027 Jiayuan Ren (expected)
2028 Rumeng Liu (expected)
2029 Zhenglei Wu (expected)

Undergraduate Class Adviser

Geochemistry Strong Foundation Program, Class 1, Cohort 2022, School of Earth and Space Sciences, Peking University (recognized as an "Outstanding Class Collective" of the School for the 2022–2023 academic year)

11. Professional Activities

(1) Committee

2025–2030, Vice Chairman, Committee on Palaeogeographic Reconstruction Technology and Big Data, International Society of Palaeogeography

2022–present, Committee Member, Anthropocene Research Society, Geological Society of China

(2) Editorial Board

2021–2027, ***National Science Review***, Editorial Group Member
2026–2027, ***Science Bulletin***, Youth Editorial Board Member
2025–present, ***Continent and Life Evolution***, Youth Editorial Board Member
2024–present, ***Acta Sedimentologica Sinica***, Youth Editorial Board Member
2023, ***Quaternary Sciences***, Invited Editorial Member

(3) Conference Chair and Convener

2027 European Geosciences Union (EGU) General Assembly, Vienna, Austria, **Co-Convener**, Session *CL4.1: Orbital forcing of global and regional events throughout Earth history*

2026 The 32nd Annual Meeting of the Palaeontological Society of China (PSC), Xi'an, China, **Convener**, *S19: High-Resolution Integrated Stratigraphy and Geochronology*
2026 The 7th International Palaeogeography Conference, Mendoza, Argentina, **Convener**, *T7-7: Astronomical forcing of palaeoclimate and palaeoenvironmental systems*
2026 The 5th International Stratigraphic Congress (STRATI), Kunshan, Jiangsu, China, **Co-Convener**, *G12: Cyclostratigraphy and its applications in geochronology and paleoclimatology*
2026 European Geosciences Union (EGU) General Assembly, Vienna, Austria, **Co-Convener**, Session *CL4.1: Orbital forcing of global and regional events throughout Earth history*
2026 The 9th Chinese Conference on Artificial Intelligence and Big Data for Earth Sciences & The 4th Summit Forum on Spatial Earth Big Data, Hefei, Anhui, China, **Co-Convener**, Topic: *Big Data, Artificial Intelligence, and the Evolution of Habitable/Early Earth*
2025 The 3rd National Symposium on Data-Driven Earth Science Development, Zhuhai, Guangdong, China, **Co-Convener**, Topic *24: AI Paradigms in Big Data Earth Science Research*
2025 Conference on Earth System Science (Shanghai), **Convener**, *Basin Sedimentary Systems and the Evolution of the Global Water Cycle*
2025 Conference on Earth System Science (Shanghai), **Co-Convener**, *Paleoocean and Paleoclimate of the Greenhouse Earth*
2025 Conference on Earth System Science (Shanghai), **Co-Convener**, *Pre-Cenozoic Orbital Forcing and Global Change*
2025 8th National Sedimentology Conference, **Co-Convener**, *High-Resolution Chronostratigraphy of Fine-Grained Sediments*
2025 8th National Sedimentology Conference, **Co-Convener**, *Orbitally Driven Environmental and Climate Evolution*
2024 AGU Fall Meeting, **Co-Chair**, *PP53G: Astronomical Forcing of Earth's Paleoclimate System II (Oral)*
2024 AGU Fall Meeting, **Co-Chair**, *PP51B: Astronomical Forcing of Earth's Paleoclimate System I (Poster)*
2024 2nd National Symposium on Data-Driven Geosciences, **Co-Convener**, *Session 1: Big Data and Artificial Intelligence Paradigm in Earth Science Research*
2024 The 6th International Conference of Palaeogeography, **Co-Chair**, *T7-8: Astronomical Forcing of Paleoclimate and Paleoenvironmental Systems*
2023 AGU Fall Meeting, **Co-Chair**, *PP01: Online Poster Session for Paleoceanography and Paleoclimatology I*
2023 AGU Fall Meeting, **Chair and Primary Convener**, *PP21D: Cyclostratigraphy and Astronomical Forcing of Earth's Paleoclimate System I* Poster
2023 AGU Fall Meeting, **Chair and Primary Convener**, *PP24B: Cyclostratigraphy and Astronomical Forcing of Earth's Paleoclimate System II* Oral
2023 Conference on Earth System Science (Shanghai), **Chair and Convener**, Session 36: Long-period orbital forcing of Earth system evolution
2023 Conference on Earth System Science (Shanghai), **Chair and Convener**, Session 37: Paleoenvironmental reconstruction during periods of global warming
2022 AGU Fall Meeting, **Chair and Primary Convener**, *PP11C: Cyclostratigraphy and Astronomical Forcing of Earth's Paleoclimate System I Online Poster Discussion*
2022 AGU Fall Meeting, **Chair and Primary Convener**, *PP13B: Cyclostratigraphy and Astronomical Forcing of Earth's Paleoclimate System II Oral*
2022 AGU Fall Meeting, **Chair and Primary Convener**, *PP25D: Cyclostratigraphy and Astronomical Forcing of Earth's Paleoclimate System III Poster*
2022 AGU Fall Meeting, **Co-Chair and Co-Convener**, *PP14A: Ocean Deoxygenation During Past Hyperthermals I Online Poster Discussion*

2022 AGU Fall Meeting, **Co-Chair and Co-Convener**, *PP22D: Ocean Deoxygenation During Past Hyperthermals II Poster*
2021 AGU Fall Meeting, **Primary Convener**, *PP006: Cyclostratigraphy and Astronomical Forcing of Earth's Paleoclimate System*
2021 GSA Annual Meeting, **Co-Convener**, *T171: Data-Driven Approaches Deciphering Water and Carbon Cycles in Earth-Surface Systems*
2021 Virtual European Geosciences Union General Assembly, **Co-Convener**, *CL1.8/SSP2.4: Climate response to orbital forcing*
2020 AGU Fall Meeting, **Chair and Primary Convener**, *Topical Session: Astronomical Forcing and Past Climate Cycles*
2020 Goldschmidt Conference, **Co-Chair and Co-Convener**, *Topical Session: New Developments in Deep-Time Paleoceanography: Geochemical Proxies, Cyclostratigraphy & Data Analysis*
2019 AGU Fall Meeting, **Chair and Primary Convener**, *Topical Session: Cyclostratigraphy and Astronomical Forcing of Past Climates*
2019 AGU Fall Meeting, **Chair and Primary Convener**, *Topical Session: Chrono-stratigraphy using Magnetic Methods*
2018 AGU Fall Meeting, **Chair and Primary Convener**, *Topical Session: Cyclostratigraphy and Astrochronology in Deep Time*
2018 International Symposium on Deep-time Environmental & Climatic Extremes and Biotic Responses, Wuhan, China, **Co-Chair**, *Session: Triassic*
2018 European Geosciences Union General Assembly, **Co-Convener**, *Session CL1.31: Climate response to orbital forcing*
2017 AGU Fall Meeting, **Co-Chair and Co-Convener**, *Topical Session PP42B: Cyclostratigraphy and Astronomical Forcing of Past Climates*
2017 GSA Annual Meeting, **Co-Chair and Co-Convener**, *Topical Session T47: Recent Developments in Cyclostratigraphy*

(4) Proposal Referee

China, National Natural Science Foundation of China (NSFC)
USA, National Science Foundation (NSF)
UK, Natural Environment Research Council (NERC)
Germany, Deutsche Forschungsgemeinschaft (German Research Foundation, DFG)
Poland, Polish National Science Centre (NCN)
Netherlands, Dutch Research Council (NWO)

(5) Journal Referee (130+ review reports)

Publons: https://publons.com/researcher/1455928/mingsong-li/peer-review/

(1) Science, (1) Science Advances, (3) Nature Communications, (3) PNAS, (31) Palaeogeography, Palaeoclimatology, Palaeoecology, (11) Global and Planetary Change, (8) Paleoceanography and Paleoclimatology, (7) Earth and Planetary Science Letters, (7) Earth-Science Reviews, (5) Journal of Asian Earth Sciences, (4) GSA Bulletin, (4) Marine and Petroleum Geology, (3) Climate of the Past, (3) Computers & Geosciences, (3) Cretaceous Research, (3) Geological Journal, (3) Geology, (3) Geophysical Research Letters, (2) Scientific Reports, (3) Quaternary Science Reviews, (3) Science China Earth Sciences, (2) Gondwana Research, (2) Journal of King Saud University - Science, (2) Science Bulletin, (2) Terra Nova, (1) Arabian Journal of Geosciences, (1) Frontiers in Earth Science, (1) Geoscience Frontiers, (1) Journal of Earth Science, (1) Journal of Geophysical Research: Atmospheres, (1) Journal of Ocean University of China, (1) Journal of the Geological Society, (1) MethodsX, (1) Paleoceanography, (1) Precambrian Research, (1) Sedimentology

(6) Judge

2022, 5th "Academic Star" Competition for Graduate Students, School of Earth and Space Sciences, Peking University

2021, 4th "Academic Star" Competition for Graduate Students, School of Earth and Space Sciences, Peking University
2019 Outstanding Student Presentation Award (OSPA), AGU
2018 Outstanding Student Presentation Award (OSPA), AGU
2017 10th Annual Postdoc Research Exhibition, Penn State

(7) Science Communication

Acycle Time-Series Analysis WeChat Official Account (Owner, approximately 7,000 subscribers)

12. Selected Media Coverage

2026 Peking University Homepage, Peking University News, and the International Organization Affairs Platform of the China Association for International Science and Technology Cooperation. "Mingsong Li of the School of Earth and Space Sciences, Peking University, Receives the 4th International Sun Shu Award" / "Mingsong Li of Peking University Wins an International Award!"
https://news.pku.edu.cn/xwzh/dda45c73785b41b78522f75364203159.htm

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